

LENNOX SAMA KEEBLE

Curriculum Vitae

Last updated: August 16, 2026.

Education

Wake Forest University

Doctor of Philosophy (Ph.D.) in Physics. Supervisor: Professor Alejandro Cárdenas-Avendaño.

2026 - 2031

Winston-Salem, USA

University of Cambridge

Master of Advanced Study (MASt) in Theoretical Physics (Part III of the Mathematical Tripos).

2025 - 2026

Cambridge, UK

Supervisor: Dr. Lorenzo Gavassino.

Princeton University

Bachelor of Arts (B.A.) in Physics. Supervisor: Professor Frans Pretorius.

2020 - 2025

Princeton, USA

Honours, Scholarships & Awards

- Cambridge Trust Master's Studentship, University of Cambridge, 2025.
- Academic Futures Graduate Scholarship, University of Oxford, 2025. (Declined.)
- Allen G. Shenstone Prize in Physics, Princeton University, Department of Physics, 2025.
- Manfred Pyka Memorial Prize, Princeton University, Department of Physics, 2022.

Publication Summary

- **5** (submitted/accepted) refereed papers; **4** as (co-)lead author.
- **Papers by research area:** relativistic hydrodynamics (**2**); black hole imaging (**2**); extreme mass-ratio inspirals (**1**).
- Links to academic metrics: [Google Scholar](#), [INSPIRE](#).

Invited Talks, Conferences & Workshops

Bold **numbers** indicate a contributed talk; bold **text** indicates an invited talk.

8. IGWN-PI School on Gravitational Waves, Perimeter Institute for Theoretical Physics, 2026.
7. Relativistic Astrophysics Summer School, Caltech, 2025.
6. Nuclear Physics of MultiMessenger Mergers (NP3M) Summer School, Indiana University, 2025.
- 5. Los Alamos Astrophysics Distinguished Seminar Series, Center for Nonlinear Studies, Los Alamos National Laboratory, 2025.**
4. Black Hole Mimickers: From Theory to Observation, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2025.
3. CU²MIP Undergraduate Conference, College Park, USA, 2024.
- 2. American Physical Society April Meeting, 2024.**
1. Taking it to the extreme: Symmetries and dynamics of extremal black holes, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2024.

Professional Experience

Investment Associate Intern

Bridgewater Associates

Summer 2024

Westport, Connecticut, USA

Summer Business Analyst

McKinsey & Company

Summer 2022

San Francisco, California, USA

Miscellaneous

Programming Languages: julia., Python, Mathematica & Slurm.

Web Links: [Webpage](#), [GitHub](#).

List of Publications

[†] indicates first-authored papers or alphabetical ordering.

Refereed (submitted/accepted)

- [5] *Radial oscillations of viscous stars.*
[†]**L. S. Keeble** and J. Redondo-Yuste.
[Physical Review D 113, 124058 \(2026\)](#) [arXiv: 2603.23622 \(gr-qc\)](#).
- [4] *First-order viscous relativistic hydrodynamics on the two-sphere.*
[†]**L. S. Keeble** and F. Pretorius.
[Physical Review D 112, 124034 \(2025\)](#) [arXiv: 2508.20998 \(gr-qc\)](#).
- [3] *Estimating high-order time derivatives of Kerr orbital functionals.*
[†]**L. S. Keeble** and A. Cárdenas-Avendaño.
[Physical Review D 112, 084048 \(2025\)](#) [arXiv: 2508.08888 \(gr-qc\)](#).
- [2] *Inferring black hole spin from interferometric measurements of the first photon ring: A geometric approach.*
[†]**L. S. Keeble**, A. Cárdenas-Avendaño, and D. C. M. Palumbo.
[Physical Review D 111, 103042 \(2025\)](#) [arXiv: 2502.20312 \(astro-ph.HE\)](#).
- [1] *Assessing the impact of instrument noise and astrophysical fluctuations on measurements of the first black hole photon ring.*
A. Cárdenas-Avendaño, **L. Keeble**, and A. Lupsasca.
[Physical Review D 109, 124052 \(2024\)](#) [arXiv: 2404.01083 \(gr-qc\)](#).

White Papers and Conference Proceedings

- [1] *The Black Hole Explorer: photon ring science, detection, and shape measurement.*
A. Lupsasca, A. Cárdenas-Avendaño, D. C. M. Palumbo, M. D. Johnson, S. E. Gralla, D. P. Marrone, P. Galison, P. Tiede, and **L. Keeble**.
[Proc. SPIE Int. Soc. Opt. Eng. 13092, 130926Q \(2024\)](#) [arXiv: 2406.09498 \(gr-qc\)](#).